Thermodynamic State Vector Inventory Discrepancy Evaluator Core Helper

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Sprint 064

Abstract

In computational ecosystems and thermodynamic simulation engines like the **Web of Life**, maintaining strict adherence to conservation laws and systemic homeostasis is paramount. Sprint 064 introduces the **Thermodynamic State Vector Inventory Discrepancy Evaluator Core Helper** (`src/thermodynamics/state_validator.ts`). This module establishes isolated, pure mathematical comparison routines that evaluate absolute inventory discrepancies against individual elemental and energy tolerances. By formalizing state verification through rigorous mathematical delta bounds, the engine ensures the enforcement of the First Law of Thermodynamics (mass and energy conservation) and bounds entropic dissipation under the Second Law. All source code and verification suites are maintained in the official repository: `https://github.com/pascalranoroarijaona/WebOfLife`.

1 Introduction and Thermodynamic Foundations

The Web of Life simulation architecture models biogeochemical cycles (carbon, nitrogen, phosphorus, water) and thermal energy flows as high-dimensional state vectors. As monad processing pipelines execute state transformations, continuous invariant checking is necessary to prevent spontaneous energy creation or mass leakage.

The core objective of Sprint 064 is to provide a robust, deterministic validation utility (`ThermodynamicStateValidator`) that compares an expected thermodynamic state vector $\mathbf{v}_{\text{exp}}$ against an actual observed state vector $\mathbf{v}_{\text{act}}$ across all inventory dimensions.

2 Mathematical Formalism & Discrepancy Evaluation

Let an inventory vector $\mathbf{v}$ be represented as a mapping of elemental stocks and energy/entropy metrics $k \in K$:

$$\mathbf{v} = \{k_1 : x_1, k_2 : x_2, \ldots, k_n : x_n\} \tag{1}$$

For expected and actual vectors, the absolute discrepancy Δ_k for each inventory key k is calculated as:

$$\Delta_k = |x_{\text{act},k} - x_{\text{exp},k}| \tag{2}$$

Each component k possesses an associated tolerance τ_k derived from `ThermodynamicToleranceConfig` (defaulting to $\tau_{\text{default}} = 10^{-6}$):

$$\text{IsValid}_k = \begin{cases} \text{true}, & \text{if } \Delta_k \leq \tau_k \\ \text{false}, & \text{if } \Delta_k > \tau_k \end{cases} \tag{3}$$

The global validity of the state transition is evaluated as the logical conjunction of all element-wise validities:

$$\text{isValid}_{\text{global}} = \bigcap_{k \in K} \text{IsValid}_k \tag{4}$$

3 Implementation Architecture

The TypeScript module `src/thermodynamics/state_validator.ts` implements these equations as a pure, side-effect-free evaluation class:

```
import { ThermodynamicStateVector } from './state_vector';
import { ThermodynamicToleranceConfig } from './types';

export interface DiscrepancyDetail {
  readonly expected: number;
  readonly actual: number;
  readonly delta: number;
  readonly tolerance: number;
}

export interface DiscrepancyResult {
  readonly isValid: boolean;
  readonly discrepancies: Record<string, DiscrepancyDetail>;
  readonly maxDelta: number;
}

export class ThermodynamicStateValidator {
  constructor(private readonly defaultTolerance: number = 1e-6) {}

  public evaluateDiscrepancy(
    expected: ThermodynamicStateVector,
    actual: ThermodynamicStateVector,
    tolerances?: ThermodynamicToleranceConfig
  ): DiscrepancyResult {
    const discrepancies: Record<string, DiscrepancyDetail> = {};
    let maxDelta = 0;
    let isValid = true;

    const keys = new Set([
      ...Object.keys(expected.inventory || {}),
      ...Object.keys(actual.inventory || {})
    ]);

    for (const key of keys) {
      const expVal = expected.inventory[key] ?? 0;
      const actVal = actual.inventory[key] ?? 0;
      const delta = Math.abs(actVal - expVal);

      const tolerance = tolerances?.getElementTolerance(key) ?? this.defaultTolerance;

      if (delta > maxDelta) {
        maxDelta = delta;
      }

      const elementValid = delta <= tolerance;
      if (!elementValid) {
        isValid = false;
```

```
      }

      discrepancies[key] = {
        expected: expVal,
        actual: actVal,
        delta,
        tolerance
      };
    }

    return { isValid, discrepancies, maxDelta };
  }
}
```

4 Conservation Verification Matrix

Table 1 outlines the core biogeochemical and thermal transitions monitored by the validator.

Process / State Transition	Conserved Quantity	Tolerance Bound (τ)	Validation
Isothermal Carbon Fixation	Total Carbon (C_{tot})	$\leq 10^{-6}$ kg	Pass if ΔC
Hydrological Flux (Evapotranspiration)	Water Mass (H_2O_{tot})	$\leq 10^{-5}$ kg	Pass if ΔH
Nitrogen Mineralization	Total Nitrogen (N_{tot})	$\leq 10^{-6}$ kg	Pass if ΔN
Thermal Dissipation (2nd Law)	Internal Energy / Entropy	$\leq 10^{-4}$ J	Pass if ΔE

Table 1: Conservation verification matrix across ecological state transformations.

5 Conclusion

Sprint 064 establishes a vital mathematical safeguard within the Web of Life thermodynamic engine. By isolating discrepancy evaluation into a pure core helper, the framework guarantees reliable invariant checking across complex ecological simulations.

Repository Reference: `https://github.com/pascalranoroarijaona/WebOfLife`